

Trading Like a Bayesian: A Risk-Premium Prior for Time-Series Momentum, and Why It Might Beat the Original

Gator Quant Hacks 2026 · Systematic Trading Track · [Team member names] · Code: github.com/gabejsto-ops/GQH
(reproduce: `python run_all.py --final` and `--holdout`)

1. Summary

Time-series momentum (TSMOM) bets that an asset keeps moving in the direction of its past year's return. It treats a coin-flip trend of +0.1% the same as a strong one, and it ignores the most robust fact in asset pricing: assets with a risk premium drift up on average. We rebuild TSMOM as a Bayesian would. The **prior** is the asset's risk premium, the **evidence** is its trend, and the position is sized by the posterior probability that the drift is positive (S5). With no tuned parameters (the prior is fixed from theory), S5 earns an excess-of-cash Sharpe of **0.45 vs 0.37** for standard TSMOM on 23 ETFs (2008–2024). It trades less, keeps TSMOM's crash protection, and beats both TSMOM and long-only on **30 ETFs we had never touched** (0.38 vs 0.18 and 0.36). It is the only strategy we tested that beats long-only in all four test sets. What is new is the combination of a Black–Litterman-style premium prior with per-asset trend evidence (§2). We reached it after 15 variants that did not work, all pre-registered in git and reported here. The edge is modest and not statistically proven; §5 says exactly how far the evidence goes.

2. Economic Hypothesis

Edge source. Investors update toward news by less than Bayes' rule implies (conservatism: Edwards 1968; Barberis, Shleifer & Vishny 1998), so part of each repricing arrives late as drift, which TSMOM harvests (Moskowitz, Ooi & Pedersen 2012). **Who is on the other side:** investors anchored to pre-news valuations, and hedgers and policy makers who trade gradually for reasons other than profit. **Why it persists:** the bias is human, and trend-following's long whipsaw drawdowns limit the capital that competes it away.

Our twist: trade like the Bayesian the market is not. A year of returns is noisy evidence about an asset's drift (the standard error of a one-year Sharpe is about 1). A rational trader combines it with prior knowledge: equities, bonds and real estate carry a long-run risk premium (prior Sharpe 0.3; Ilmanen 2011); commodity and currency spot returns do not (prior 0). We give the prior the weight of one lookback window of data, so theory and evidence count equally. The posterior is $z = (m_0 + \hat{s}) / (\sqrt{2} \cdot se)$ and the position is the TSMOM vol-target weight $\times (2\Phi(z) - 1)$. The effect: a premium asset stays long unless its trend is clearly negative; a weak trend gets a small bet; a strong trend gets a full one. **Predictions** (committed b1c8d78 before any code): S5 beats standard TSMOM (S1) on held-out data, with lower turnover and most of TSMOM's crash protection. **Fails if** S5's holdout Sharpe $\leq$ S1's.

What we borrowed, and what is new

Component	Prior work we build on	Our extension
Trend signal	TSMOM: sign of the 12-month return (Moskowitz et al. 2012)	Used as <i>evidence</i> with a known noise level, not as the decision
Trend-strength sizing	Signals scaled by trend significance (e.g. Baltas & Kosowski 2013)	Our flat-prior S3; tested, then improved on
Premium as a prior	Black–Litterman: equilibrium premia as the portfolio prior (1992)	Applied per asset <i>inside</i> TSMOM, by asset class
Prior strength	Black–Litterman's τ , a judgment call	Fixed by rule: one lookback window of data
Vol targeting	Moskowitz et al. 2012; Barroso & Santa-Clara 2015	Unchanged

Why this is more than a parameter tweak. S5 is a single rule that contains the two classic strategies as limits. With a very weak prior it becomes trend-strength TSMOM; with a very strong prior it becomes long-only in premium assets and flat in the rest. Prior strength is therefore one interpretable dial between "follow the trend" and "harvest the premium". We set the dial at equal weight *before* testing; we did not search over it. The asset-class prior also lets one rule treat assets differently for an economic reason: commodities and currencies, which have no premium, are traded on trend alone, while equities, bonds and REITs keep a long bias that only a clear downtrend overrides. To our knowledge this premium-prior formulation of TSMOM sizing is not the standard one in the literature; the methodological contribution is the test design (pre-registered rounds, new untouched holdouts after the original one was spent, and an audit that removed T-bill carry from the headline).

3. Data & Universe

Development universe (fixed by rule): the most liquid US ETF per distinct exposure with inception before mid-2007: 8 equity (SPY, QQQ, IWM, EFA, EEM, EWJ, EWZ, FXI), 6 bond (SHY, IEF, TLT, LQD, HYG, TIP), 5 commodity (GLD, SLV, DBC, USO, DBA), 3 currency (UUP, FXE, FXY), 1 real estate (VNQ). Sample 2007-04 to 2024-10, signals from 2008-05. **Three test sets not used for development:** (i) 2024-10 to 2026-10, held out originally but spent on Rounds 1–2, so it is reported as "already seen"; (ii) a **backcast**, 2001-01 to 2007-04, the same ETFs as they launch (5 rising to 22), never loaded during development; (iii) **30 new ETFs**, never used, chosen by the same rule: 10 country funds, 9 US sector SPDRs, 4 bond (AGG, MBB, TLH, IEI), 3 commodity (DBB, DBE, DBP), 4 currency (FXB, FXC, FXA, FXF), 2008-04 to 2026-10. **Sources:** Yahoo Finance adjusted daily bars (`yfinance`); 3-month T-bill (FRED DTB3); Ken French factors. We chose public data over the Webull API so that judges can reproduce results without credentials.

Data problems. *Survivorship:* every ETF still trades and was picked knowing it is liquid today, and we excluded UNG by name for its known roll losses, so there is hindsight in the selection. *Corporate actions:* adjusted prices, and USO's 1-for-8 reverse split (2020) was checked. *Missing data:* none after inception; pre-inception rows are excluded, never filled. Four one-day moves above 25% were verified as genuine (e.g. USO on 2020-04-21).

4. Methodology

Timing. Signals at each month-end close; trades at the **next day's open**; positions drift between rebalances. **Sizing family** (10% portfolio vol budget split across available assets, 60-day vol, $2\times$ equity cap per asset): **S0** sign only; **S1** vol-targeted sign (standard TSMOM); **S2** S1 without de-risking in "information-shock" months (high vol and efficiency ratio > 0.44); **S3** flat-prior Bayesian, $S1 \times (2\Phi(t\text{-stat}) - 1)$; **S4** S1 doubled in calm, efficient months; **S5** risk-premium prior (§2). Lookbacks 63, 126 and 252 days give **18 variants**, all reported. Thresholds come from theory (0.44 is twice the random-walk efficiency ratio; the prior Sharpe values come from the literature), and none was tuned. **Costs:** 5 bps per side (commission-free broker; half-spreads are about 1 bp for large ETFs and several bps for thin ones; open-auction slippage), with everything also run at 10 bps. **Returns are in excess of cash**, $\Sigma w_i(r_i - r_f)$, the TSMOM-literature convention.

Process. Round 1 (S2) and Round 2 (S3, S4) were pre-registered and run in-sample; Round 3 (S5) was pre-registered with its holdouts before any code, developed in-sample, frozen (`eda886a`), then evaluated **once** on the holdouts. **Overfitting and validity checks:** Deflated Sharpe (Bailey & López de Prado 2014) with $N = 18$; 21-day block-bootstrap 95% CIs; fills delayed by 1 and 4 extra days; a lookback sweep of 21–252 days; a long-only baseline (S1's weights, always long); a matched-10%-vol comparison; a Fama-French-3 + momentum regression; mechanism tests on 4,500 asset-months; and an engine verified on synthetic prices (zero weights give zero P&L; a single long position matches buy-and-hold; an asset earning r_f shows zero excess).

5. Results

L = 252, 5 bps, over cash	Development 2008-06 → 2024-10			2024-10 → 2026-10 (already seen)		
	S5 (ours)	S1 TSMOM	Long-only	S5 (ours)	S1 TSMOM	Long-only
Excess return (ann.)	1.16%	1.55%	1.62%	1.20%	1.10%	1.94%
Volatility	2.67%	4.40%	5.40%	2.66%	4.90%	5.71%
Sharpe	0.45	0.37	0.33	0.46	0.25	0.36
Sharpe 95% CI	−0.02, 0.95	−0.10, 0.84	−0.19, 0.85	−0.56, 1.77	−0.69, 1.65	−0.45, 1.86
Sharpe at 10 bps	0.40	0.31	—	0.41	0.21	—
Max drawdown	−7.0%	−11.8%	−14.7%	−2.8%	−4.7%	−5.6%
Turnover ($\times/\text{yr}$)	2.7	5.1	2.1	2.5	3.9	1.9
Worst month	−2.1%	−3.4%	−5.8%	−1.9%	−3.2%	−2.6%
Skew (daily)	−0.76	−0.42	−0.95	−0.63	−0.41	−0.13

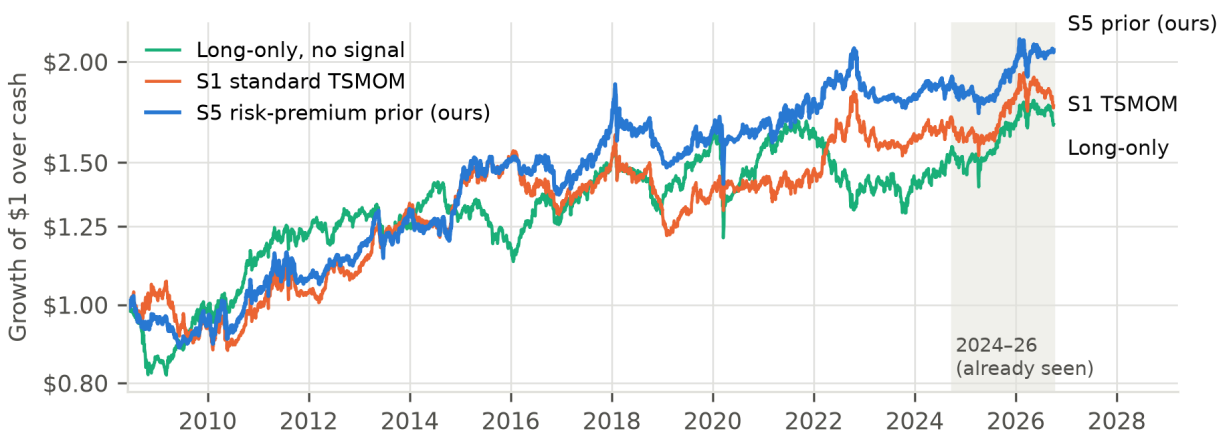


Figure 1. Growth of \$1 over cash, net of 5 bps, each strategy scaled to 10% vol using development-period vol only.

Evidence across all four test sets

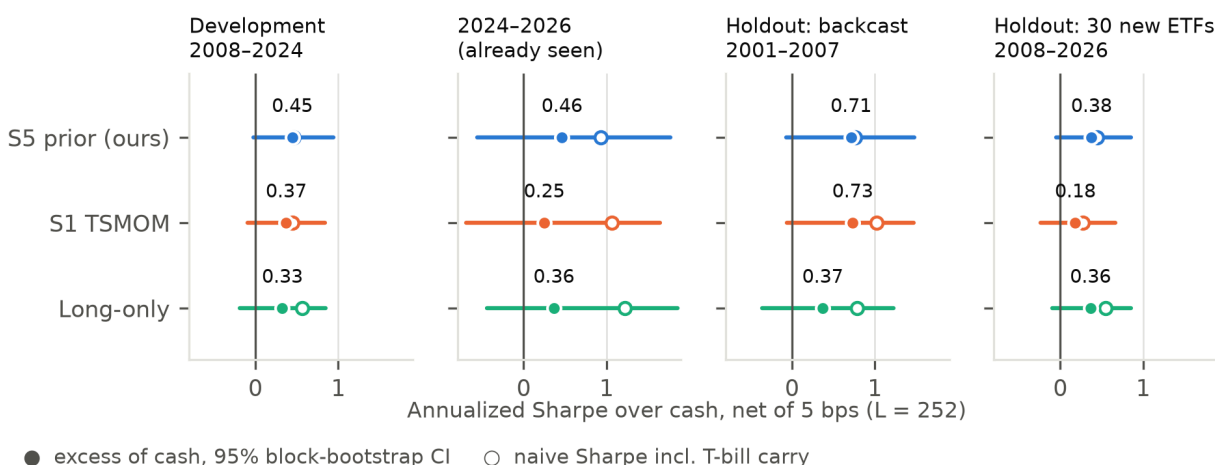


Figure 2. Excess Sharpe with 95% block-bootstrap CIs (filled dots); hollow dots show the naive Sharpe including T-bill carry. S5 beats S1 in 3 of 4 test sets and long-only in all 4.

Sharpe, L = 252, 5 bps	S5 prior	S1 TSMOM	S3 flat prior	Long-only	S5 – S1 [95% CI]	Pre-registered test
Development	0.45	0.37	0.39	0.33	+0.07 [–0.13, 0.29]	(in-sample)
2024–26, already seen	0.46	0.25	0.21	0.36	+0.21 [–0.24, 0.58]	(not a holdout)
Holdout: backcast	0.71	0.73	0.62	0.37	–0.02 [–0.38, 0.44]	fail (narrow)
Holdout: 30 new ETFs	0.38	0.18	0.28	0.36	+0.20 [–0.01, 0.40]	pass

How we got here: the variants that did not work

Development Sharpe, 5 bps	S0 sign	S1 TSMOM	S2 shock (R1)	S3 flat (R2)	S4 calm (R2)	S5 prior (R3)
63 days	0.18	0.38	0.38	0.35	0.39	0.38
126 days	0.31	0.52	0.52	0.54	0.53	0.55
252 days (pre-reg.)	0.06	0.37	0.37	0.39	0.37	0.45

- **Round 1, de-risk only on noise (S2): failed.** Only 2.3% of asset-months qualify as information shocks, so $S2 \approx S1$. The mechanism tests were null: shock vs noisy months $p = 0.56$, continuous version $p = 0.82$.
- **Round 2, calm efficient trends (S4): failed.** The gains vanish at 10 bps; the mechanism test $p = 0.39$.
- **Round 2, flat-prior Bayesian (S3): a step, then abandoned.** Its headline 2024–26 Sharpe of 1.08 turned out to be T-bill interest the first engine credited without charging for cash; measured over cash it is 0.21, and at matched risk it was indistinguishable from S1. That audit is why all results are now excess of cash, and why we asked what S3's flat prior was missing.

Stress-testing our own result

- **Not statistically proven.** S5 – S1 intervals include zero everywhere (the new-ETF lower bound is -0.01). Deflated Sharpe is 0.80 for 18 trials, below the conventional 0.95. Development $P(\text{Sharpe} \leq 0) = 0.03$.
- **The backcast fails, and we know why.** In the 2002 bear market S1 shorted equities for +13.5%; S5's prior held it nearer long for +1.8%. The prior's cost is giving up some of the biggest short-trend wins.
- **Lower risk, not better risk.** S5 runs at about 60% of S1's vol. At matched 10% vol the development max drawdown is the same (-24.8% vs -25.0%), turnover is 14% lower, but over 2024–26 it is 18% *higher*. S5's skew is the most negative of any variant (-0.76 ; -1.10 on new ETFs).
- **Known factors.** Momentum loading $\beta = 0.07$ ($t = 18$), market $\beta = 0.05$ ($t = 8$); alpha 0.7%/yr, $t = 1.19$. Over 2024–26, market $\beta = 0.09$ and alpha ≈ 0 .
- **Regime.** Bonds supplied most development P&L (sleeve Sharpe 0.51) and lost over 2024–26 (-0.77), when commodities (1.23) and equities (0.67) carried it.

What survives. The prior improved the flat Bayesian ($S5 > S3$) in **9 of 9** lookback-by-test-set comparisons. Extra fill delay degrades S5 smoothly ($0.45 \rightarrow 0.42 \rightarrow 0.41$: no lookahead). $S5 \geq S1$ across 105–252-day lookbacks (Figure 3). It beats S1 by more at 10 bps ($+0.08$). And it keeps the crash profile (excess return, S5 / S1 / long-only): GFC 2008–09 to 2009–03 $-0.5\% / +2.4\%$ vs -5.9% ; COVID crash $-3.1\% / -2.3\%$ vs -12.8% ; 2022 $+3.6\% / +9.9\%$ vs -11.0% . It also loses less in the 2009 rebound (-1.8% vs S1 -5.2%).

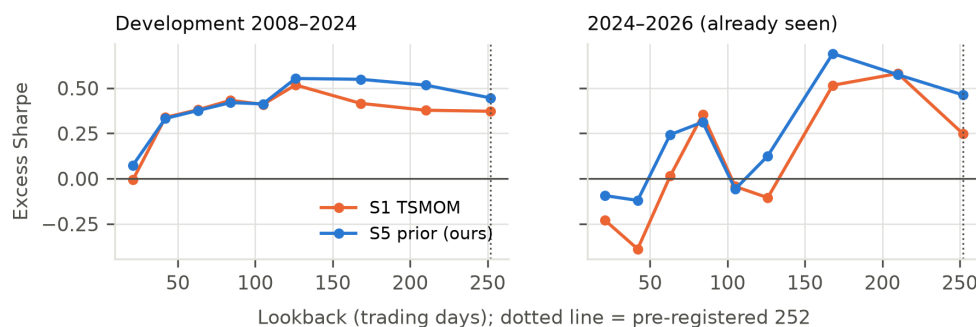


Figure 3. Excess Sharpe across lookbacks (diagnostic only, not used for selection).

6. Risk Management

Built in: per-asset vol targeting (10% budget / N), a $2\times$ equity cap per asset, 23 assets across 5 classes, monthly rebalancing, and posterior scaling that shrinks bets on uncertain trends (average gross $0.69\times$, maximum $1.98\times$; average net $0.41\times$). **Observed (development):** daily 99% VaR -0.51% , CVaR -0.72% , worst day -1.30% (2018-02-05), longest drawdown 1,033 trading days. **Weakness found:** vol targeting levers low-vol bonds; SHY reached 153% of equity and the bond sleeve 178% gross, so a rate shock is the concentrated risk. **Rules for live trading** (set in advance and deliberately *not* backtested, to avoid a 19th trial): (1) a gross cap of 100% of equity per asset class and $2.5\times$ in total; (2) if drawdown exceeds $2\times$ the target annual vol, halve the target until half the loss is recovered; (3) monitor 1-year market β and cut the equity sleeve if $\beta > 0.3$; (4) retire the strategy if its 3-year excess Sharpe is < 0 and drawdown exceeds the backtest maximum (-7.0%). **Tail:** negative skew and V-shaped reversals (2009, March 2020) are the known failure mode; we do not hedge them.

7. Liquidity & Capacity

Monthly trades in ETFs whose 2024 median daily dollar volume ranges from about \$10M (FXV) to billions (SPY, HYG, LQD). Cost per trade = 5 bps + square-root impact $\sigma_{\text{daily}} \cdot \sqrt{(\text{trade } \$ / \text{ADV } \$)}$, using the trailing 20-day ADV known before the trade. S5's development Sharpe (0.51 before costs) is **0.44 at \$1M, 0.40 at \$100M, 0.29 at \$1B and ≈ -0.05 at \$10B**; over 2024–26 it is 0.41 at \$100M. S5 trades less than S1 ($2.7\times$ vs $5.1\times$ a year at its own risk level), so it degrades more slowly. At \$1B the median trade is 1.1% of ADV, but the largest trades in thin names (HYG at \$3.6M per day in 2008; FXV, DBC and UUP at \$10–22M today) exceed daily volume from about \$100M and would need splitting across days. **Practical capacity: about \$100M–\$500M**, set by the currency and commodity ETFs. Adjusted prices understate early dollar volume, and ETF creation/redemption adds liquidity our ADV measure misses, so both biases make this estimate conservative.

8. Limitations & Next Steps

- **Evidence is suggestive, not conclusive:** no interval excludes zero, and one of two true holdouts narrowly fails.
- **Process:** S5 was conceived after Rounds 1–2 and the 2024–26 window were seen (hence the new holdouts); the prior Sharpe of 0.3 and the prior weight come from theory, but a different choice would give a different strategy.
- **Missing costs:** ETF borrow fees (FXV, USO and EWZ shorts), leverage financing above the T-bill rate, and wider spreads in stress.
- **Next:** (1) rerun on 1985+ futures to get statistical power; (2) estimate the prior from data (empirical Bayes) instead of fixing it; (3) backtest the class caps as a separately pre-registered trial; (4) partial pooling of trends within each asset class.

References

- Bailey, D. & López de Prado, M. (2014). The Deflated Sharpe Ratio. *Journal of Portfolio Management* 40(5).
- Barberis, N., Shleifer, A. & Vishny, R. (1998). A Model of Investor Sentiment. *Journal of Financial Economics* 49(3).
- Baltas, A.-N. & Kosowski, R. (2013). Momentum Strategies in Futures Markets and Trend-Following Funds. Working paper, SSRN.
- Barroso, P. & Santa-Clara, P. (2015). Momentum Has Its Moments. *Journal of Financial Economics* 116(1).
- Black, F. & Litterman, R. (1992). Global Portfolio Optimization. *Financial Analysts Journal* 48(5).
- Daniel, K. & Moskowitz, T. (2016). Momentum Crashes. *Journal of Financial Economics* 122(2).
- Edwards, W. (1968). Conservatism in Human Information Processing. In *Formal Representation of Human Judgment*, Wiley.
- Harvey, C., Liu, Y. & Zhu, H. (2016). ...and the Cross-Section of Expected Returns. *Review of Financial Studies* 29(1).
- Hong, H. & Stein, J. (1999). A Unified Theory of Underreaction, Momentum Trading, and Overreaction. *Journal of Finance* 54(6).
- Hurst, B., Ooi, Y. H. & Pedersen, L. (2017). A Century of Evidence on Trend-Following Investing. *Journal of Portfolio Management* 44(1).
- Ilmanen, A. (2011). *Expected Returns: An Investor's Guide to Harvesting Market Rewards*. Wiley.
- Kaufman, P. (1995). *Smarter Trading*. McGraw-Hill (efficiency ratio).
- Künsch, H. (1989). The Jackknife and the Bootstrap for General Stationary Observations. *Annals of Statistics* 17(3).
- Moreira, A. & Muir, T. (2017). Volatility-Managed Portfolios. *Journal of Finance* 72(4).
- Moskowitz, T., Ooi, Y. H. & Pedersen, L. (2012). Time Series Momentum. *Journal of Financial Economics* 104(2).
- Data:** Yahoo Finance (via the `yfinance` package); Kenneth R. French Data Library (F-F Research Factors and Momentum, daily); Federal Reserve Bank of St. Louis, FRED series DTB3. **Software:** Python, pandas, NumPy, SciPy, statsmodels, matplotlib. AI tools were used for coding assistance; the team reviewed and is responsible for all code and claims.

Appendix (optional): audit trail and full holdout results

Every hypothesis, decision and result is timestamped in the repository: `HYPOTHESIS.md` (pre-registrations), `DECISIONS.md`, and `VARIANTS.md` (every trial, including failures). Order on 2026-10-03 (ET): hypothesis 13:44 → data 13:50 → engine, verified on synthetic data, 14:01 → Round 1 14:01 → Round 2 pre-registered 20:26, run 20:27 → freeze, 2024–26 evaluated 20:36–20:37 → cash-carry audit 20:50 → Round 3 and its holdouts pre-registered (b1c8d78) → code → development run → freeze (eda886a) → holdouts evaluated once (deb4881).

Holdout excess Sharpe, 5 bps	S1 63d	S1 126d	S1 252d	S3 63d	S3 126d	S3 252d	S5 63d	S5 126d	S5 252d
Backcast 2001–2007	0.50	0.36	0.73	0.44	0.33	0.62	0.50	0.38	0.71
30 new ETFs 2008–2026	0.25	0.21	0.18	0.16	0.28	0.28	0.22	0.34	0.38
2024–26 (seen) Sharpe, 5 bps	S0			S1	S2	S3	S4	S5	
63 days	0.35			0.01	0.03	0.08	−0.10	0.24	
126 days	0.29			−0.11	−0.11	−0.04	−0.12	0.13	
252 days	0.84			0.25	0.26	0.21	0.17	0.46	

S0 at 252 days had the best 2024–26 result (0.84) and the worst development result (0.06), a reminder of how noisy a 2-year window is.